ESWAR RITHIK M

Generative AI Engineer | Python | LLMs | RAG | LangChain | Azure OpenAI | FastAPI

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PROFESSIONAL SUMMARY

MCA graduate interested in Generative AI, LLMs, and AI application development. Hands-on experience building RAG-based applications using Python, LangChain, Azure OpenAI, Azure AI Search, and FastAPI. Developed projects in document Q&A, natural language to SQL analytics, and AI-powered code assessment. Eager to apply my skills and continue learning while building practical AI solutions.

TECHNICAL SKILLS

Programming Languages: Python, SQL, JavaScript

Generative AI & LLMs: LLMs, Generative AI, RAG Architecture, Prompt Engineering, Grounded Answer Generation, AI Agents, Function Calling, NLP

Retrieval & Vector Search: Embeddings, Document Ingestion, Chunking, Retrieval, Semantic Search, Hybrid Search, Reranking, ChromaDB, FAISS, Azure AI Search

Frameworks & AI Platforms: LangChain, LangGraph, Azure OpenAI, OpenAI API, Google Gemini AI, Hugging Face, Sentence Transformers

Backend & Databases: FastAPI, REST APIs, PostgreSQL, SQLite

Machine Learning & Data: Machine Learning, Deep Learning, Transformers, Scikit-learn, Pandas, NumPy

Web Technologies: React.js, Streamlit, HTML5, CSS3, Material UI, Responsive Web Design

Tools & Cloud: Git, GitHub, Docker, Azure, VS Code, Postman, Swagger UI

INTERNSHIP EXPERIENCE

Generative AI / AI Developer Intern Apr 2026 – Sep 2026 (6 Months) | Chennai

Centaur IT Billing

- Created a healthcare document Q&A application over 50+ documents using Retrieval-Augmented Generation (RAG) with LangChain, Azure OpenAI, and Azure AI Search.
- Developed an end-to-end pipeline that handles ingestion, chunking, metadata enrichment, embedding generation, vector indexing, retrieval, and answer generation.
- Implemented hybrid search and reranking to make retrieval more relevant for healthcare and claims-related questions.
- Applied prompt engineering with Azure OpenAI to produce context-aware answers grounded in retrieved enterprise documents.
- Integrated FastAPI with the retrieval and generation components to deliver an AI workflow.
- Collaborated with cross-functional teams to test and refine the application, using their feedback to improve answer accuracy and user experience.

Python Developer Intern May 2025 – Jul 2025 (3 Months) | Chennai

Entrust Software & IT Solutions

- Developed Python scripts and modules using OOP, data structures, file handling, and exception handling.
- Used Pandas and NumPy to manipulate, clean, and analyze structured datasets.
- Built and integrated REST APIs and managed database connectivity using SQL.
- Wrote and debugged Python scripts and modules, adding exception handling and basic unit tests to improve reliability.
- Used Git and GitHub (branching, committing, merging) in an agile team, participated in code reviews and daily stand-ups, and helped write internal documentation for scripts and API endpoints.

PROJECT EXPERIENCE

AI Document Assistant — RAG-Based Document Assistant GitHub

Tech Stack: *Python, LangChain, ChromaDB, Google Gemini, Hugging Face Embeddings, Streamlit*

- Created a RAG-based question-answering system that allows users to ask questions about PDF documents using natural language, combining Google Gemini with Hugging Face embeddings and semantic search to understand and retrieve information.
- Designed a document processing pipeline that handles ingestion, chunking, embedding generation, and vector storage in ChromaDB for retrieval.
- Built a user interface with Streamlit where users can upload documents and ask questions in a chat-like format.

QueryCraft AI — GenAI-Powered Natural Language-to-SQL Analytics Platform GitHub

Tech Stack: *Python, FastAPI, React.js, PostgreSQL, Google Gemini AI, REST APIs, Swagger UI*

- Developed a full-stack analytics platform that turns natural language questions into optimized SQL queries using Google Gemini AI, enabling non-technical users to perform data analysis without writing code.
- Automated importing of CSV and Excel files with dynamic schema detection and automatic creation of PostgreSQL tables.
- Built a FastAPI backend with 5+ REST API endpoints to support the React.js frontend, enabling real-time query execution and result visualization, with API documentation provided through Swagger UI.

CodeMentor AI — AI-Powered Software Project Assessment Platform GitHub

Tech Stack: *Python, FastAPI, React.js, Google Gemini AI, GitHub API*

- Engineered an AI-powered platform that reviews GitHub repositories and produces project summaries, architecture insights, and suggestions for improving code quality using Google Gemini AI.
- Integrated the GitHub API to collect repository metadata, commit history, and file structures across multiple repositories to enable automated project evaluation.
- Delivered a responsive full-stack application that generates dynamic PDF reports for users to download and analyze.

CERTIFICATIONS

- NPTEL - Introduction to Machine Learning
- Python Training, IIT Bombay (2025)
- PET-26 Paper Presentation - AI-Based Nutrition Planning and Monitoring for Children with Disabilities
- ICTDT-24 Paper Presentation - E-Commerce Technologies
- Brainovision International Student Workshop (2024)

KEY ACHIEVEMENTS

- Built 3 AI-powered full-stack applications and completed two Python/AI internships totaling 9 months of hands-on experience.
- Developed 5+ REST APIs using FastAPI across projects.
- Built RAG-based AI solutions using LLMs, LangChain, and Azure OpenAI.

EDUCATION

Master of Computer Applications (MCA) 2024 – 2026 | Chennai

B.S. Abdur Rahman Crescent Institute of Science and Technology

CGPA: 8.8 / 10

Bachelor of Computer Applications (BCA) 2021 – 2024 | Chennai

Prince Shri Venkateshwara Arts and Science College

CGPA: 8.5 / 10

LANGUAGES KNOWN

English, Tamil